Can Weight Changes Predict Performance? A Proposed Study of Model-Edit Trajectories

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Abstract

Can weight changes predict how a model's measured capabilities change, without running the edited model? Recent work by Takehi et al. reports that a small, targeted edit to Qwen3.5-0.8B can improve a benchmark score without gradients (Takehi et al., 2026). That result motivates this question but does not show that effects can be predicted across edits or models. We propose to test whether weight-only features and edit histories predict benchmark changes. We will also test transfer to unseen edit families and compare an LLM reader with a trained predictor. A planned alignment sub-study will compare instruction-tuned, directionally ablated, and Heretic-edited models on capability, refusal, and KL-to-base measures. These outcomes remain unknown; we will test the hypotheses on a controlled edit zoo. The study will focus on prediction and auditing, not gradient-free optimization or a new safety-editing method.

1 Introduction

Can a model's weight changes tell us how its measured capabilities will change before we run the edited model? The question matters when evaluating a large collection of fine-tunes or interventions becomes expensive. A proposed weight-space predictor would estimate the direction and approximate size of each performance change without executing every candidate. Such estimates would not replace evaluation. They could help prioritize edits for full benchmark runs and flag capability costs that might be missed when measuring only the intended behavior.

Our starting point is Takehi et al.'s report that editing 0.016% of Qwen3.5-0.8B's weights raised its MATH-500 score from 7.9% to 48.1% without gradients (Takehi et al., 2026). This is a measured effect of a weight edit. It does not show that the effect can be predicted from a delta or that it generalizes to other edit families. We propose a narrower test: given a base model and a sequence of weight edits, can a predictor estimate changes on a fixed benchmark battery without new forward passes? We will represent each transition as $\Delta\theta_t = \theta_t - \theta_{t-1}$. The target will be a vector of benchmark changes, not a single score or a natural-language account of the edit.

We propose four hypotheses and their decision criteria. H1 predicts the sign and rough magnitude of per-benchmark changes on held-out edits to the same base model. It will meet its criterion only if per-benchmark Spearman $\rho \geq 0.6$ and it beats both the weight-statistics baseline (StatNN) and the norm-of-delta baseline. H1 will be falsified if $\rho < 0.3$ on all benchmarks after honest tuning. H2 tests whether a stateful sequence predictor improves on a stateless per-edit predictor. Its target is at least 15% relative MAE reduction on trajectories of length at least 4. H2 will be falsified if the stateless model is within noise. H3 tests transfer from abliteration and random-direction edits to unseen LoRA fine-tunes. Its target is transfer $\rho \geq 0.5$. H3 will be falsified if transfer $\rho < 0.2$. These are planned decision rules, not results. Under H4, we will compare a frontier LLM and a trained predictor given the same delta as text, summary statistics, and SVD components. We expect the trained predictor to be more accurate on in-distribution edits. We expect the LLM reader to be more robust to novel edit families. We will report both directions on the same held-out sets.

We also propose an alignment-edit study to test prediction across model sizes. We will compare base-instruct checkpoints, classic directional ablations, and Heretic edits on Qwen3.5-0.8B, 4B, and 9B and on Qwen3.8-27B. We will measure capability benchmarks, refusal and over-refusal rates, and KL divergence from the base. Separately, we will test whether a predictor can identify the edit family from a delta and estimate associated capability changes. A refusal marker does not fully measure harmfulness. A small KL divergence does not establish that task accuracy is preserved. The study will audit existing edits rather than propose a new abliteration method.

We propose three components. First, we will build a controlled set of single edits and multi-edit trajectories with paired benchmark outcomes. We will score predictions under splits by edit family and seed. Second, we will compare weight statistics and canonicalized low-rank features with stateless, sequential, and JEPA-style predictors. We will also evaluate an LLM reader. These comparisons will test whether sequence context or learned representations add predictive value beyond per-edit summaries. Third, we will apply the same prediction framework to alignment edits and report capability, refusal, and KL measures together. The proposed edit zoo will initially center on Qwen models, so conclusions will depend on the edit families and evaluation battery it covers. We report no experimental outcomes here, and we do not claim that weight-only prediction can replace benchmark evaluation.

2 Related Work

Learning from model weights. Weight-space learning covers work on weight geometry, on learned representations of parameters, and on generating new parameters (Han et al., 2026b). Our question is a narrower one: predicting measured performance changes from the weight difference between successive checkpoints. W2T is a close precedent for the representation side of this problem. It canonicalizes LoRA updates through QR decomposition and SVD, tokenizes the resulting components, and processes them with rank-level and position-level Transformers. On its ARC-LoRA benchmark, W2T reports test MAE 0.32 and Spearman 95.35% (Han et al., 2026a). Those numbers concern LoRA performance prediction on that benchmark. We cannot read off their performance on merged edits, multi-edit trajectories, or the benchmarks in our proposed study. We therefore treat canonicalized low-rank features as a candidate representation, not as evidence that the same accuracy will carry over.

Several neighboring lines predict performance by running the model first. Delta Activations represents fine-tuned models by measuring how their internal activations change relative to a base model (Xu et al., 2025). Fast probing reads intermediate hidden activations to predict downstream performance. It reports AUROC above 0.75 and cuts evaluation latency from about an hour to three minutes on OLMo3-7B-base (Liu et al., 2026). TuneAhead pairs dataset descriptors with short-probe features and reports a 1.47 percentage-point RMSE over 370 held-out Qwen2.5-7B-Instruct runs (Luo et al., 2026). These methods motivate a cost-aware comparison. Their inputs include activation or probe information. Our proposed setting will use weight deltas and their summaries without additional forward passes. Comparisons will therefore need to account for different inputs and compute budgets.

Edit trajectories and predictive representations. Weight changes may carry information about an edit's effect. A later edit may also depend on earlier edits. The task-vector study reports a twenty-fold collapse in interference on identical prompts under a training-format wrapper, but no collapse under an untrained chat template (+12.5 points) (Zhu et al., 2026). This finding motivates recording evaluation formats and modeling trajectory history. It also cautions against treating a merge as context-free addition. A separate study across CIFAR training and Pythia checkpoints reports localized, time-varying pockets of predictable short-horizon dynamics in bulk parameters (Wang et al., 2026). Leap+Verify proposes speculative future-weight predictions and accepts them only after validation against a held-out loss criterion (McEntire, 2026). That proposal does not establish that our performance predictor will reduce training time.

We will test sequence models because each observed edit arrives in context. An edit can change the starting point for the next one. Mamba-3 pairs a more expressive state-space recurrence with complex-valued state updates and a multi-input, multi-output formulation (Lahoti et al., 2026). That description is architectural; it does not demonstrate an advantage on model-edit trajectories. Under H2, we will test whether a stateful

sequence predictor improves held-out performance estimates over a stateless per-edit model. A separate JEPA-style model will predict a representation of the next weight state from the prior state and an edit descriptor. The JEPAWG work on physical parameterized systems omits the VICReg covariance term from its main-text configuration, which allows a lower-dimensional latent representation (Göbel et al., 2026). ResidualPred is reported to bias transformer predictor attention toward identity at initialization. It also improves effective rank and downstream accuracy in the reported tabular and image experiments (Costa et al., 2026). These findings will inform our design, but neither demonstrates transfer across language-model edit families. A separate analysis distinguishes four levels of JEPA/HMM correspondence and gives sufficient conditions for exact sequence-level HMM equivalence (Huang, 2026).

Alignment edits and evaluation limits. Refusal-related weight edits provide a useful stress test because refusal and task capability may change together. Heretic combines configurable modifiers with Optuna TPE optimization, and its default modifier is Arbitrary-Rank Ablation (Weidmann, 2026). We plan to compare this family with classic directional ablation. We will measure capability rather than assume it is preserved. A separate analysis of Heretic's default refusal score notes that it matches any of 33 case-insensitive strings within the first 100 response tokens; the score alone does not measure harmfulness (Sharma, 2026). We will report refusal measures alongside capability and distributional change. We will interpret keyword-based rates narrowly.

Prior work shows why safety or collateral effects should not be inferred from one intervention or proxy. Surgical Refusal Ablation orthogonalizes a refusal vector against protected concept atoms. Across five models it reports low refusal rates, alongside perplexity and first-token KL changes. Those measurements do not establish universal preservation of task accuracy (Cristofano, 2026). Directional ablation on GLM-5.3-Flash (320B) left measurable refusal after every tested edit, including tested subspaces of ranks 1–12 (Shi et al., 2026). This result applies to the tested edits; it does not show that directional ablation cannot change refusal in general. We will use such findings to motivate several endpoints in the proposed alignment sub-study, not as predicted results for Qwen. Our proposed contribution is an empirical test of whether performance changes can be predicted from deltas and trajectories. We will treat prior findings as evidence about their tested settings. The proposed predictor will not replace benchmark evaluation.

3 Method

Let θ_0 be a base model and let e_t be an edit that produces θ_t from θ_{t-1}. The step delta is $\Delta\theta_t = \theta_t - \theta_{t-1}$. Let $y_t \in \mathbb{R}^K$ hold the benchmark outcomes of θ_t. We propose predicting $\Delta y_t = y_t - y_{t-1}$ from the weights, and predicting y_t along edit sequences. For deployed bf16 weights W and a proposed update Δ, decision D1 defines $\Delta_{\text{eff}} = \text{bf16}(W + \Delta) - W$. Features should describe this effective change. Rounding statistics record the lost entries and the norms before and after rounding. An fp32 merge is a separate experiment.

The map ϕ_{stats} collects per-tensor Frobenius and relative norms and the cosine similarity of previous and new weights. It also records the mean, standard deviation and maximum absolute value, the positive and nonzero fractions, and the leading singular values with spectral entropy and stable rank. The spectral summaries use the retained singular values. The map ϕ_{svd} expresses each matrix update as singular triplets. For LoRA factors it applies QR decompositions followed by an SVD of the small core, which removes dependence on the choice of factorization. Dense updates use a truncated SVD. The sign of each pair is fixed by the largest absolute component of the left vector. Each rank token holds its singular value and fixed random projections of its left and right vectors. Degenerate singular subspaces may still admit multiple bases.

We propose six predictor families. M1 is a stateless tensor-token Transformer. M2 is a state-space sequence model that takes pooled delta features and edit descriptors. M3 predicts a subsequent weight embedding with a JEPA objective and a variance regularizer. M4 is an LLM reader given textual statistics and singular components. M5 is a causal trajectory Transformer. M6 pretrains a tabular JEPA on statistics; its encoder is then frozen, and a supervised head is trained on top. Delta-norm linear regression, LightGBM on the statistics, and an MLP on SVD features serve as baselines. We plan these comparisons. No predictor results are reported here.

Family	Production	Smoke	Unclassified
Random direction	150	0	3
Abliteration	200	0	5
Heretic ARA	0	0	1
LoRA SFT	120	0	1
Task vector	80	0	0
Chain steps	523	0	20

Table 1: Local `data/zoo/qwen3.5-0.8b/manifest.jsonl`, read 4 November 2026. Counts are rows by recorded eligibility; unclassified denotes legacy rows without that field.

We will evaluate family-held-out, seed-held-out and ancestry-grouped splits. Ancestry grouping uses the full manifest, so an intermediate, unevaluated edit cannot separate a descendant from its root. Separate family and seed splits do not by themselves guarantee ancestry isolation. For each benchmark we will report Spearman correlation and mean absolute error, with bootstrap confidence intervals over edits. Trajectory comparisons will resample independent ancestry groups. The preregistered sequence test sets a bar of at least 15% relative MAE reduction on trajectories of length at least four.

4 Edit zoo

The builder implements six families. Random-direction ablation projects selected weight matrices away from sampled directions. Projected abliteration takes its refusal directions from differences in mean activations on harmful and harmless prompts. Its Heretic-parity classic kernel applies $W' = W - w_\ell v(v^\top W)$ with no row normalization. The layer weight w_ℓ interpolates linearly from a peak to a floor inside a finite radius and is zero outside it. Directions may be selected or interpolated across residual layers. This is one specific projected variant. It does not use Heretic's default row normalization.

Heretic ARA generates an adapter from Optuna trials scored on refusal and KL. A configured KL constraint restricts which trials are eligible. Heretic describes ARA as its default modifier, and it optimizes with Optuna TPE (Weidmann, 2026). LoRA supervised fine-tuning trains adapters on GSM8K or Alpaca-style data with sampled training parameters. Task-vector entries scale an eligible parent's delta relative to the pinned base. Chains store cumulative snapshots relative to the base. Random-direction and abliteration steps modify the accumulated state. Adapter steps are optimized against the base and then added to that state. To get step deltas, consecutive snapshots must be differenced. A snapshot must never be applied again to its parent.

An entry ID hashes the family, the parameters, the seed, the pinned base revision and the parent. A separate reuse key records further provenance: the kernel, the dtype, the refusal fingerprint and the chain prefix. Any mismatch blocks silent reuse. Eligibility separates production entries from smoke entries and legacy unclassified ones. Smoke status propagates through descendants. A task-vector parent must have a compatible base and provenance, and it is normally a production entry.

Table 1 counts manifest rows from the local build as of writing, leaving out the header lines. Chain rows count cumulative steps, not independent trajectories. The manifest is a build artefact. It does not show that every row has a completed evaluation.

5 Experiments

The subject is Qwen3.5-0.8B with thinking enabled. Fixed subsets cover GSM8K, MATH-500, IFEval, MMLU-Pro and HellaSwag. We also measure WikiText perplexity, refusal on harmful prompts, over-refusal on harmless prompts, and KL to the base. Loop rate is the fraction of generated responses that reach the token limit. Refusal keyword matching is a surface measure. It does not establish harmfulness.

The proposed fast screening tier uses 50 GSM8K, 25 MATH-500, 25 IFEval, 100 MMLU-Pro and 100 HellaSwag items, plus 50 prompts each for refusal, over-refusal and KL. Generation is capped at 1024

Metric	Ours, 4k (%)	Blog, 32k (%)
GSM8K	15.5	13.8
MATH-500	6.0	7.9
IFEval	42.0	21.0
MMLU-Pro	20.3	—
HellaSwag	51.3	—
Loop rate	90.2	93.0
Refusal rate	99.0	—
KL to base	0.000	—

Table 2: Base reproduction, run `p2/base-final-4096`, and blog values recorded in `docs/RESULTS.md`. Our loop rate pools GSM8K, MATH-500 and IFEval; the blog value is MATH-500 only. KL is a self-check, not a percentage.

tokens. The cheap tier runs the larger fixed battery with a 4096-token limit. A faithful 32768-token tier is reserved for the base and selected entries. Fast scores are the proposed predictor targets, and cheap scores will calibrate a subset. We plan to start with a clean timing measurement and a stratified production subset, and to extend coverage after that. Sampling settings and fixed item IDs are recorded in the evaluation configuration.

Table 2 gives the completed base run alongside the blog comparison recorded in `docs/RESULTS.md`. The budgets and loop denominators differ, so these figures are not matched estimates. The 32k rerun was interrupted when the GPU ran out of memory, and the IFEval differences still need checker reconciliation. Local work runs on an RTX PRO 4500 with 32 GB. The Heretic ARA family runs on Lambda GPU compute: one NVIDIA A100-SXM4 40 GB instance at $1.99/h.

References

José Lucas De Melo Costa, Seong Woo Ahn, Fabrice Popineau, Arpad Rimmel, and Bich-Liên Doan. Drive vs. Decay: On the Training Dynamics of Joint-Embedding Predictive Architectures. *arXiv preprint arXiv:2610.02344*, 2026.

Tony Cristofano. Surgical Refusal Ablation: Disentangling Safety from Intelligence via Concept-Guided Spectral Cleaning. *arXiv preprint arXiv:2601.08489*, 2026.

Tobias Göbel, Julian R. Ebelt, Zier Mensch, Mathis Gerdes, and Miranda C. N. Cheng. Weight-Space Physics: Interpretable Hypernetworks for Lattice Quantum Field Theories. *arXiv preprint arXiv:2607.07127*, 2026.

Xiaolong Han, Ferrante Neri, Zijian Jiang, Fang Wu, Yanfang Ye, Lu Yin, and Zehong Wang. W2T: LoRA Weights Already Know What They Can Do. *arXiv preprint arXiv:2603.15990*, 2026a.

Xiaolong Han, Zehong Wang, Bo Zhao, Binchi Zhang, Jundong Li, Damian Borth, Rose Yu, Haggai Maron, Yanfang Ye, Lu Yin, and Ferrante Neri. A Survey of Weight Space Learning: Understanding, Representation, and Generation. *arXiv preprint arXiv:2603.10090*, 2026b.

Yongchao Huang. Your Probabilistic JEPA Is Secretly a Hidden Markov Model: A State-Space Interpretation of Joint-Embedding Predictive Learning. *arXiv preprint arXiv:2608.13621*, 2026.

Aakash Lahoti, Kevin Y. Li, Berlin Chen, Caitlin Wang, Aviv Bick, J. Zico Kolter, Tri Dao, and Albert Gu. Mamba-3: Improved Sequence Modeling using State Space Principles. *arXiv preprint arXiv:2603.15569*, 2026.

Zhichen Liu, Tianle Lun, Zhibin Wen, Hao An, Yulin Ou, Jianhui Xu, Hao Zhang, Wenyi Fang, Yang Zheng, and Yang Xu. Fast and Accurate Probing of In-Training LLMs' Downstream Performances. *arXiv preprint arXiv:2604.01025*, 2026.

Yuxiang Luo, Haonan Long, Chen Wang, Qiqi Duan, Xiaotian Lin, Yanwei Xu, Yuyu Luo, Weikai Yang, and Nan Tang. TuneAhead: Predicting Fine-tuning Performance Before Full Training Begins. *arXiv preprint arXiv:2606.17660*, 2026.

Jeremy McEntire. Leap+Verify: Regime-Adaptive Speculative Weight Prediction for Accelerating Neural Network Training. *arXiv preprint arXiv:2602.19580*, 2026.

Parminder Kumar Sharma. Heretic automates stripping refusals from open-weight models. Its headline score counts 33 words, not harm, 2026. URL `https://www.pk-sharma.com/briefing/heretic-open-weight-refusal-removal-controls-outside-the-model`. P.K. Sharma briefing, 21 September 2026.

Yi Shi, Tanyu Chen, and Kai Shen. How Fragile Is Safety Alignment at Frontier Scale? A Single-Direction Attack on a 320B MoE. *arXiv preprint arXiv:2609.09793*, 2026.

Rikiya Takehi, Jayden Teoh, Alex Zhang, and Omar Khattab. Sparks of Weight-Space Programming: LLMs can natively read & write neural network weights, 2026. URL `https://rikiyat.github.io/blog/weight-space-programming/`. Blog (MIT CSAIL), 2 Nov 2026.

Fanqi Wang, Weisheng Tang, and Hairong Qi. Measuring Structured Predictability in Neural Training Dynamics: A Cross-Regime Study. *arXiv preprint arXiv:2608.15483*, 2026.

Philipp Emanuel Weidmann. Heretic, 2026. URL `https://github.com/p-e-w/heretic`. GitHub: p-e-w/heretic (README accessed 3 November 2026).

Zhiqiu Xu, Amish Sethi, Mayur Naik, and Ser-Nam Lim. Delta Activations: A Representation for Finetuned Large Language Models. *arXiv preprint arXiv:2509.04442*, 2025.

Chencheng Zhu, Xiaoyang Li, and Taotao Cai. When Do Task Vectors Interfere? Mapping the Validity Boundaries of Weight-Space Composition. *arXiv preprint arXiv:2608.09490*, 2026.